SAMPREETH AVVARI

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SUMMARY

AI Engineer who ships LLM and ML products to production: RAG systems with grounded citations, agentic multimodal pipelines, and forward-deployed work built beside clinicians and the C-suite. Shipped 5 production systems solo under HIPAA-eligible controls.

EDUCATION

MS, Computer Engineering, New York University, NY (CGPA 3.84/4) **Aug 2023 - May 2025**
BTech, Computer Science and Engineering, JNTU, India (CGPA 3.92/4) **Jul 2017 - Jun 2021**

SKILLS

AI: LLM product development, GenAI, prompt engineering, context engineering, JSON Schema contracts, agentic workflows (tool use, multi-step orchestration), evals, rubrics, guardrails, RAG (pgvector, Milvus), embeddings, fine-tuning, RLHF (GRPO/PPO), QLoRA, PyTorch, Transformers, Vertex AI Gemini, Llama, LangChain
Languages: Python, TypeScript, JavaScript, Go, Java, SQL, C++; FastAPI, Flask, Node.js, REST, GraphQL, microservices, React, Next.js
Cloud: PostgreSQL, pgvector, MongoDB, Redis, BigQuery, Kafka; GCP (Cloud Run, Pub/Sub), AWS (S3, RDS, MSK), Docker, Kubernetes, Terraform, MLflow, Airflow, CI/CD, monitoring (Prometheus, Grafana, OpenTelemetry)
Delivery: LLM products from prototype to production, agent design, eval-driven development, guardrails and safety, model selection, fast iteration with stakeholders; Google Gen AI Leader

WORK EXPERIENCE

AI Engineer | Hybridge Implants, New York **Sep 2025 - Present**

- Partnered with clinicians and the C-suite to ship 5 production ML and LLM systems solo, taking models from prototype to production and replacing a $98K vendor quote. Systems span RAG, agentic diagnosis, call QA, and medical-imaging CV across 3 clinic locations.

Doctor & NPC Coach, consultation and call-QA platform (Doc Blog, NPC Blog)

- Shipped a Vertex AI Gemini pipeline scoring every Zoom-recorded consult and new-patient call on a clinical rubric, emailing each doctor a coaching report minutes after the call and lifting treatment acceptance 130%.
- Engineered the scorer on Gemini 2.5 Pro with JSON-schema-validated output and retry on schema failure, requiring a verbatim transcript quote behind every score and cutting hallucinated feedback 35%.
- Made the pipeline self-healing with Firestore run tracking, exponential-backoff retries, and stuck-job recovery, raising reviewed calls from near 0% to 100% across 2 Zoom organizations with zero PHI in logs.

CDF, agentic multimodal diagnosis assistant (Blog)

- Shipped a structured second read on Vertex AI Gemini 2.5 across a 3-location implant practice, reaching 90% concordance with senior dentists over 200+ cases, cutting consult prep from an hour to a 5-minute packet and cross-branch treatment-plan variance 40%.
- Architected a 6-agent pipeline fusing 5 modalities (CBCT/DICOM, TRIOS scans, 9-view photos, radiographs, intake survey) into one schema-validated JSON packet scoring caries risk, 0-100 tooth-loss, and 6-tier treatment leaning, with human-in-the-loop review.

Hybridge Enterprise Search, company-wide RAG platform (Blog)

- Delivered grounded question answering with mandatory inline citations over 100k+ indexed documents, resolving conflicting sources through an authority ladder and abstaining when evidence is thin, cutting hallucinated answers 70% at 95% faithfulness and 1.2s p95 for $300 a month on GCP. A/B evaluated against Vertex AI Search to justify the in-house build on accuracy, citation fidelity, and cost.
- Drove recall@10 to 92% with hybrid BM25-dense retrieval (gemini-embedding-001, pgvector HNSW) via Reciprocal Rank Fusion and Vertex reranking, enforcing document ACLs across 12 groups with pre-retrieval SQL filtering.

CBCT Scan Validator, artifact-detection model (Blog)

- Deployed a multi-task classifier across 13 artifact classes into the implant-planning workflow, flagging 12% of 600 monthly CBCT scans as unusable before design while holding sensitivity near 0.90.
- Benchmarked 6 architectures on a 10,000+ study CBCT corpus with PyTorch Lightning DDP on 4 A100s in fp16, shipping the top model at macro AUROC 0.93 with DVC and GCS data versioning.

Software Engineer | Shure Incorporated, India **Aug 2021 - Aug 2023**

- Engineered event-driven Flask REST APIs on AWS ingesting 500K+ daily Kafka events, cutting processing time 20% with async workers draining MSK topics into DynamoDB and S3 for downstream test analytics.
- Directed delivery across 3 release tracks with Jenkins canary deploys, cutting deploy errors 40% with 150+ pytest and Selenium tests at 90%+ coverage, and led code reviews. Parallelized the regression suite, cutting runtime 65% from an 8-hour run to under 2.5 hours.

RESEARCH EXPERIENCE

Machine Learning Researcher | New York University (Blog) **May 2024 - Sep 2025**

- Fine-tuned Llama 3.1 8B through an RLHF lifecycle (SFT then GRPO/PPO, QLoRA reward models), reaching a 67% win rate on human evaluation. Benchmarked PPO against GRPO from one frozen base and blocked length-bias reward hacking with length-matched pairs.
- Engineered a Python and Airflow ETL turning 118 Reddit CMV shards into 38K preference pairs with schema enforcement and failure alerting, tracked in WandB. Split pairs by debate thread to stop train-test leakage and trained a reward model with 71% pairwise accuracy.

PROJECTS

JobPilot, open-source (MIT) AI job-hunt autopilot (Code, Demo, Blog)

- Built JobPilot, an open-source job-hunt autopilot, AI-scoring fit and auto-tailoring an ATS-clean resume per match for under $10 monthly.
- Designed 7 agentic patterns, adopted by 12 peers: an LLM-as-judge scorer, JSON-schema output contracts, a self-refining rewrite loop, a RAG assistant, multi-step orchestration, tool-use across Gmail, Sheets, and Drive, and guardrails that never auto-send.

Loan Radar, self-retraining loan-default scoring platform (Code, Blog)

- Architected Loan Radar, a self-retraining loan-default platform on Terraform Kubernetes (Ray) with 3 FastAPI microservices, Docker, and MLflow-tracked CI/CD for rollback-safe predictions.